



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2024

Bachelor in Computer Applications

Course Title: Data Structure and Algorithms

Code No: CACS 201

Semester: III

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

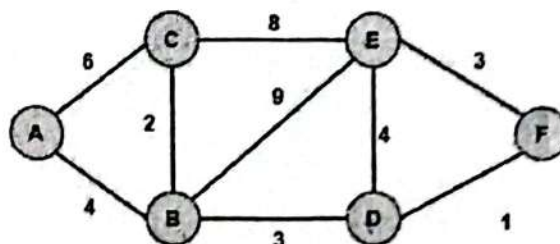
2. Differentiate between stack and queue. What are the general applications of a stack? [2+3]
3. What is a linked list? How doubly linked list is different from circular linked list? Explain with example. [1+4]
4. What is recursion and recursive function? Write a recursive function to compute Fibonacci number. [2+3]
5. How does collision occur during hashing? Explain any two hashing functions. [3+2]
6. What is an AVL tree? Create an AVL tree from the following data: [1+4]
18, 12, 14, 8, 25, 31, 24, 27
7. What are deterministic and non-deterministic algorithms? Explain the use of Big Oh notation to measure the complexity of an algorithm with an example. [2+3]
8. Implement the quick sort to sort the following data items: [5]
12, 1, 14, 7, 2, 10, 4, 7, 22, 6, 15

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What are the differences between linear queue and circular queue? Write an algorithm to enqueue and dequeue data elements in a circular queue. [4+3+3]
10. What is B-tree? How insertions and deletions of elements can be done in a B-tree. [2+8]
11. Explain the different ways to represent a graph. For the following graph use Prim's algorithm to find a minimum spanning tree starting from the node 'A'. [5+5]





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Bachelor in Computer Applications

Course Title: OOP in Java

Code No: CACS 204

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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is constructor? Explain the role of constructor in Java class by considering a suitable example and also list the type of constructor. [1+3+1]
3. Define inheritance. Explain types of inheritance with suitable example. [1+4]
4. Define Package. Explain steps in creation and implementation of package with example. [1+4]
5. What is multithreading? How does Java support inter-thread communication? Explain with example. [1+4]
6. How to handle multiple catch blocks for a nested try block? Explain with an example.
7. What is difference between String and String buffer and explain 3 methods of String and String Buffer class with example. [2+3]
8. Write short notes on: (any two)
 - i. JDBC Driver
 - ii. Access Modifiers
 - iii. JDK, JRE and JVM

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Define array and multi dimensional array. Write a program to read two m x n matrices, perform multiplication operation and store result in third matrix. [3+7]
10. Write a program to create the following simple GUI based application. If user press the Submit button, your program should store the information in a file named "exam.txt" only when he accepts the terms and condition otherwise it should display a message "please accept the terms and condition first". [6+3+1]



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Bachelor in Computer Applications

Course Title: Probability and Statistics

Code No: CAST 202

Semester: III

Full Marks: 60

Pass Marks: 24

Time: 3 hours

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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Describe the scope and limitations of Statistics.
3. What do you mean by statistics? The following table represents the marks of Probability and Statistics of 100 students.

Marks	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
No. of students	12	16	35	24	13

Find the mean, median and standard deviation of all 100 students.

4. Define correlation. From the following data on marks of 10 students in the two subjects, calculate the Karl Pearson's coefficient of correlation and interpret the result:

Maths	55	70	40	30	90	80	60	80	90	80
Basic Statistics	65	40	30	50	60	70	50	50	60	70

5. Define regression. The following table gives the age of the computers of a certain company and annual maintenance costs:

Age of computers (years)	2	4	6	8	10
Maintenance costs(Rs.00)	10	15	22	32	46

- i. Obtain the regression equation for cost related to age.
- ii. Estimate the cost of maintenance for 10 yrs old computer.
- iii. Interpret the slope.
6. Define Poisson distribution. In certain factory timing out optical lenses, there is a small chance, $1/500$ for any lens to be defective. The lenses are supplied in a packet of 10 each. What is the probability that a packet will contain; (i) No defective lens, (ii) At least one defective lenses, and, (iii) At most two defective lenses.

$p(0)$



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Bachelor in Computer Applications
Course Title: System Analysis and Design
Code No: CACS 203
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define system. Explain components of system including its characteristics. [1+2+2]
3. Define case tool. Explain different types of case tool that can be used in different phases of SDLC. [1+4]
4. Define information system planning. How is top-down planning approach different from bottom-up planning approach? Explain. [1+4]
5. Define JAD? List out different contemporary requirements determining techniques and explain them in comprehensive way. [1+4]
6. Define data modeling. Explain logical data model in detail. [1+4]
7. What are the various types of menu design which can be adopted to meet the system complexity and usability?
8. Why do software project often fails? Explain different types of software testing? [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Explain phases of SDLC, also discuss the role of Project Manager in software project development. [7+3]
10. What is process modeling? Why do we need DFD? Draw up to top level DFD for Online hotel room booking system. [1+2+7]
11. List out different activities that are associated with implementation phase. Explain each of them in detail. What are the different methods of training and supporting users? Explain. [6+4]

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2024

Bachelor in Computer Applications

Course Title: Web Technology

Code No: CACS 205

Semester: III

Full Marks: 60

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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is CSS? Explain the method of using CSS on the web page. [2+3]
3. What is XML? Write rules to create XML document with example. [1+4]
4. Explain HTML form elements with suitable example. [2+3]
5. What is XPath? Write XSLT code to display your books information (Title, price ISBN, subject code) store into XML. [1+4]
6. Why server side programming is important for web development? Explain with its major features. [5]
7. Why session is important for web development, write a server side script to create, store, retrieve and remove session and cookie. [2+3]
8. Explain multi-tier technology with its advantage and disadvantages. [2+3]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Write a server side script for implementing login. Your program should take username and password as input and redirect to home page if validated other wise prints an error try again. [10]
10. Compare XML Schema with DTD. Write XML schema to describe student as root element, name as child element with first_name, middle_name, last_name, city as child elements with its content "KTM", "PKR", "LPT" and rollno as its attribute. [3+7]
11. Create web page with following details. [10]
 - a) Create basic structure [2]
 - b) Add header section to the top of page which contain logo and search bar. [2]
 - c) Create two column layout of (25%,75%) width and put unordered list into first one and image gallery into second one. [2]
 - d) Add footer section, which includes address of company and copyright information. [2]
 - e) Also add some basic CSS using external stylesheet.[2]



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Center:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

i) What is the measurement for time complexity of an algorithm?

- a) Counting microseconds b) Counting kilobytes of algorithms
c) Counting number of key operations d) Counting number of statements

ii) Which of the following is the result of evaluation of $5\ 7\ 4 - * 8\ 4 / +$?

- a) 5 b) 8 c) 10 d) 17

iii) What is the recursive formula for post order traversal of binary tree?

- a) Left-Root-Right b) Root-Left-Right
c) Left-Right-Root d) Right-Left-Root

iv) What is the number of disk movement in TOH with 4 disks?

- a) 9 b) 14 c) 17 d) 15

v) What is the Big-Oh of best case complexity of insertion sort?

- a) $O(n)$ b) $O(n \log n)$ c) $O(1)$ d) $O(n^2)$

vi) How does the rear index incremented in circular queue?

- a) $\text{front} = (\text{rear} + 1) \% \text{SIZE}$ b) $\text{rear} = (\text{rear} + 1) \% \text{SIZE}$
c) $\text{rear} = \text{rear} + 1$ d) $\text{rear} = (\text{rear} - 1) \% \text{SIZE}$

vii) A variation of linked list in which none of the node contains NULL pointer is

- a) Singly b) Multiple c) Circular d) Doubly

viii) Which of the following data structure is used in depth first search of graph?

- a) Stack b) Queue c) Linked List d) None of the above

- ix) Which of the following is true for B-Tree of order M ?
- a) Leaf nodes should be at different level
 - b) All the key values within a node must be in descending order
 - c) Every node has at least M children
 - d) All non-leaf nodes with $M-1$ keys must have M number of children
- x) Which of the following is not a hash function?
- a) Division remainder
 - b) Folding
 - c) Chaining
 - d) Mid square



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is Data Structure? Show the status of stack converting following infix expression to post fix $P + Q - (R * S / T + U) - V * W$ [1+4]
3. Write binary search. Consider a hash table of size 10; insert the keys 62, 37, 36, 44, 67, 91 and 107 using linear probing. [2+3]
4. What are deterministic and non-deterministic algorithms? Explain greedy algorithm. [3+2]
5. Draw a BST from the string DATASTRUCTURE and traverse the tree in post order and preorder. [3+2]
6. Define circular queue? How does circular queue overcome the limitation of linear queue? Explain. [2+3]
7. What is singly linked list? Write an algorithm to add a node at the beginning and end of singly linked list. [1+4]
8. Define AVL tree. Construct AVL tree from given data set: 4, 6, 12, 9, 5, 2, 13, 8, 3, 7, 11. [2+3]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What is stack? List the applications of stack. Write an algorithm or procedure to perform PUSH and POP operation in stack. [1+2+7]
10. What is heap? Explain quick sort algorithm with Big-oh notation in best case, average case and worst case and trace it to sort the data: 8, 10, 5, 12, 14, 5, 7, 13. [2+2+6]
11. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.



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Bachelor in Computer Applications
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Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

i) Which one of the following is not a valid java bitwise operator?

a) >>

b) <<

c) >>>

d) <<<

ii) Which one of the following keyword is used to declare an exception?

a) throws

b) throw

c) try

d) catch

iii) Which of these is an incorrect array declaration?

a) `int ary[] = new int[5];`

b) `int[] ary = new int[5];`

c) `int ary = int[5] new;`

d) `int ary[]; ary = new int[5];`

iv) Which one of the following access specifier is appropriate for members of superclass to access only from subclass?

a) private

b) protected

c) public

d) default

v) Which one of the following is not a collection class defined in java?

a) Linked List

b) Hash Set

c) Tree Set

d) Graph Set

- vi) Which one of the following inheritance is best implanted using interface in java?
- a) single inheritance
 - b) multi-level inheritance
 - c) multiple inheritance
 - d) hierarchical inheritance
- vii) Which one of the following method is called only once during the run time of your applet?
- a) stop()
 - b) paint()
 - c) init()
 - d) start()
- viii) Which of these method of class String is used to compare two String objects for their equality?
- a) equals()
 - b) Equals()
 - c) is Equal()
 - d) Is Equal()
- ix) What is the default value of priority variable MIN_PRIORITY and MAX_PRIORITY?
- a) 0 & 63
 - b) 1 & 10
 - c) 0 & 1
 - d) 1 & 32
- x) Which one of the following is not java swing container?
- a) Panel
 - b) Tabbed Pane
 - c) Scroll Pane
 - d) Scroll bar



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Group B

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[6×5 = 30]

2. Define OOP. Explain features of Object Oriented Programming Language. [1 + 4]
3. Explain different types of control statements used in java. [5]
4. Define Abstract Class. Explain different types of Access controls available in java.[1 + 4]
5. Define method overriding? Write any program to implement concept of multiple inheritance in Java.
[1 + 4]
6. Why it is important to handle exception in java? Write a program to illustrate the use of exception handling. [1 + 4]
7. Define the use of **static** keyword. Write any four String methods used in java with example.
[1 + 4]
8. Define **super**, **final** and **this** keyword in java. Explain the concept of MVC in brief.
[1 + 1 + 1 + 2]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. a) Define multithreading. Write a java program to show the inter-thread communication.
[1 + 4]
b) Define Stream. Write a program in java to copy the content from one file to another.
[1 + 4]
10. a) Define Collection Class. Explain different Wrapper classes and associated methods in java.
[1 + 4]
b) Define AWT. Explain different types of Layout Managers in java. [1 + 4]
11. a) List and explain any five swing controls with their uses. [5]
b) Define JDBC. Write a program to display all records from a table of database.
[1 + 4]



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Group A

[10×1 = 10]

1. Circle (O) the correct answer.
 - i) How many types of data on the basis of sources of data collection?
 - a) 1
 - b) 2
 - c) 3
 - d) 4
 - ii) Which is more appropriate central tendency to find the average of profit?
 - a) Arithmetic mean
 - b) Median
 - c) Mode
 - d) All
 - iii) What is the range of Correlation?
 - a) 0 to ∞
 - b) $-\infty$ to ∞
 - c) -1 to 1
 - d) 0 to 1
 - iv) If $r=0.2$ then coefficient of determination implies that
 - a) 20% of total variation in dependent variable has been explained by independent variable.
 - b) 40% of total variation in dependent variable has been explained by independent variable.
 - c) 2% of total variation in dependent variable has been explained by independent variable.
 - d) 4% of total variation in dependent variable has been explained by independent variable.
 - v) What is the minimum value of Probability?
 - a) 1
 - b) 100
 - c) 0
 - d) None of above
 - vi) In case of Normal distribution
 - a) Mean > Median
 - b) Mean = Median
 - c) Mean $\leq$ Median
 - d) Mean $\geq$ Median
 - vii) The regression line of X on Y and Y on X are intersect at the point
 - a) $(\mu, 0)$
 - b) (a, b)
 - c) (X, Y)
 - d) $(\bar{X}, \bar{Y})$

viii) In case of systematic sampling

- a) sample mean is biased estimator population mean.
- b) sample mean is unbiased estimator population mean.
- c) sample mean can't estimate population mean.
- d) sample mean may equal to population mean.

ix) Mean of Chi-Square distribution with n degrees of freedom is

- a) 1
- b) 0
- c) $2n$
- d) n

x) How do you obtain degree of freedom in one-way ANOVA?

- a) $(k, n-1)$
- b) $(k, n-k)$
- c) $(k-1, n-1)$
- d) $k-1, n-k$



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Describe scope and limitation of Statistics.
3. Determine average wages from following data:

Wages	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

4. Calculate Karl Pearson's correlation coefficient from the following data:

Sales	43	41	36	34	50
Expenses	10	22	13	19	17

5. Estimate the marks in JAVA when the marks in Statistics is 65 by using following data:

Marks in Statistics	57	58	59	59	60	61	62	64
Marks in JAVA	77	78	75	78	82	82	79	81

6. Fit Binomial Distribution from the following data where $p = 0.5$

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

7. How do you determine sample size in sampling? Explain briefly.
8. Write short notes on simple random sampling.

Group C

Attempt any TWO questions.

[2×10 =

20]

9. Student's age in the regular daytime BCA program and the morning time BCA program of a campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestion, with reason, which of two groups will be easier to teach?

Regular BCA program		Morning BCA program	
Age	No. of Students	Age	No. of Students
23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	60

10. Given a normal distribution with mean 200 and s.d. 20, find the probability that
- i) $P(X > 180)$
 - ii) $P(X < 220)$
 - iii) $P(160 < X < 240)$
 - iv) $P(X > 220)$
 - v) $P(X < 180 \text{ or } X > 220)$
 - vi) 10% of the values are less than what values of X?

11. The labor productivity indexes of Nepal are recorded as below:

Sector	Year		
	2015	2016	2017
Agriculture	100	125	138
Manufacturing	100	60	53
Community and Social service	100	89	80

Does the labor productivity index vary due to the;

i) difference in the sector

ii) difference in the time period?



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Group A

Attempt all the questions.

[10×1 = 10]

2. Circle (O) the correct answer.

- i) Which of the following Information systems are aimed at improving the routine business activities on which all organizations depend?
 - a) Management Information systems
 - b) Decision support systems
 - c) Transaction Processing Systems
 - d) Executive Information
- ii) The project life cycle consists of
 - a) Understanding the scope of the project
 - b) Objectives of the project
 - c) Formulation and planning various activities
 - d) Word Processor
- iii) Which is the most important feature of spiral model?
 - a) Quality Management
 - b) Efficiency Management
 - c) Risk Management
 - d) Performance Management
- iv) includes the existing system, the proposed system, system flow charts, modular design of the system, print layout charts and data file designs.
 - a) Feasibility Report
 - b) Functional Specification Report
 - c) Design Specification Report
 - d) Terms of Reference
- v) For the best Software model suitable for the project, in which of the phase the developers decide a roadmap for project plan?
 - a) Software
 - b) System Analysis

c) Coding

d) Testing

vi) Using the approach, a new system is tested in one part of the organization before being implemented in others.

a) Direct

b) Parallel

c) phased

d) pilot

vii) extends the software beyond its original functional requirements.

a) Adaptive maintenance

b) Perfective maintenance

c) Corrective maintenance

d) Preventive maintenance

viii) Which normal form looks at removing partial dependencies?

a) First Normal Form

b) Second Normal Form

c) Third Normal Form

d) Fourth Normal Form

ix) In constructing ER diagrams, double ovals are used to denote

a) Multi-value table

b) Multi-value entity

c) Multi-value attributes

d) Multi-value key

x) Testing beyond normal operational capacity is

a) Performance testing

b) Stress testing

c) Recovery testing

d) None of the above



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[6×5 = 30]

3. When would you use agile methodologies? How is it different from waterfall approach to system development?
4. Why is project management important? Briefly explain the activities performed by the project manager during project execution.
5. List various methods of interacting with a system. Briefly explain the factors to be considered while designing a form?
6. What are the deliverables from coding and testing? Briefly explain the different approaches to installation.
7. Why is normalization required? State second normal form and explain it with a proper example.
8. Construct an E-R Diagram for football club that has a name and a ground and is made up for players. A player can play for only one club and a manager identified by his name manage a club. A footballer has a registration number, name and age. A club manager also buys players. Each club plays against other clubs in the league and matches have a date, venue and score.
9. Maintenance is an on-going process. Do you agree? Explain the process of maintaining information systems.

Group C

Attempt any TWO questions.

[2×10 = 20]

10. Develop a context diagram and top level logical DFD for the system made up of the following.

B & B is a mail-order company that distributes CDs, DVDs of music, games, movies, software at discount prices to club members.

- When an order processing clerk receives an order form, he or she verifies that the sender is a club member by checking the Member file.
- If the sender is not a member, the clerk returns the order along with a membership application form.
- If the customer is a member, the clerk verifies the order item data by checking the item file.
- Then the clerk enters the order data and saves it to the Daily Order file. The clerk also prints an invoice and shipping list for each order, which are forwarded to Order Fulfillment Department.

10. With proper reasoning, explain how CASE Tools aid in information system development? You have been hired as a system analyst in TU tech software development company and you are asked to analyze the way system works. What qualities do you need to have to analyze such type of systems?

11. a) Why software project often fails? Explain different types of software testing?
- b) List of OOAD. Differentiate between structured methodologies and object oriented methodologies.



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Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

10×1 = 10

1. Circle (O) the correct answer.

i) Which of the following provide additional information about HTML elements?

- a) Headings b) Paragraphs c) Attributes d) Styles

ii) Which of the following is NOT an inline element?

- a) b) c) <input> d) None of the above

iii) How can you make an e-mail link?

- a) b) <mail href="xxx@yyy ">
c) <mail>xxx@yyy </mail> d)

iv) For XML document to be valid?

- a) Document need to be well formed also
b) Document need not to be well formed
c) Document need to be well formed & valid
d) Document validity has no relationship with well formedness

v) W3C is stands for

- a) World Wide Web Center b) World Wide Web Company
c) World Wide Web Consortium d) World Wide Web Certificate

vi) XSL is,

- a) AN XML based language to create style sheets
b) Extensible solution language
c) A language which is not used in XML

d) Extended stylesheet language

vii) CSS uses dot (.) for,

a) Element selector

b) Class selector

c) ID selector

d) Attribute selector

viii) Namespace in XML is used to,

a) Distinguishes one XML vocabulary from another

b) Provides the spaces into name

c) Querying language

d) Make duplicate of XML validation

ix) A cookie is often used to identify a

a) User

b) Browser

c) Web page

d) Client

x) Server side scripts is

a) Visible to the end user

b) Invisible to the end user

c) Defined to user

d) None



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is CSS? Discuss types of CSS with example.
3. What is XML? Differentiate XML schema with DTD.
4. List common application of web server, Explain how session works.
5. Describe the tags and attributes for a form in HTML document?
6. What is anonymous access? Discuss about Integrated windows authentication
7. Explain tire technology with examples?
8. Write short note on following:
 - Cookies.
 - XML DOM

Group C

Attempt any TWO questions.

[2×10= 20]

9. Differentiate between tags and attributes? Write a HTML and CSS code design your curriculum vitae (CV) using different html elements like (table, image, formatting tags, links).
10. Develop a simple web page that asks the users input (name, email, phone, gender) and store into database using server side script.
11. What are the rules for creating XML document? Write a XML code to store following information about student.
 - Each student has a name, address, phone and website element.
 - Address might appears multiple times
 - Address has attribute named “type” with value permanent and temporary
 - Phone must be 10 digit.



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2019

Bachelor in Computer Applications
Course Title: Data Structures & Algorithms
Code No: CACS 201
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Center:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10 × 1 = 10]

1. Circle (O) the correct answer.

- i) What is the measurement for time complexity of an algorithm?
- a) Counting microseconds b) Counting kilobytes of algorithms
- c) Counting number of key operations d) Counting number of statements
- ii) Which of the following is the result of evaluation of $5 \ 7 \ 4 - * \ 8 \ 4 / + ?$
- a) 5 b) 8 c) 10 d) 17
- iii) What is the recursive formula for post order traversal of binary tree?
- a) Left-Root-Right b) Root-Left-Right
- c) Left-Right-Root d) Right-Left-Root
- iv) What is the number of disk movement in TOH with 4 disks?
- a) 9 b) 14 c) 17 d) 15
- v) What is the Big-Oh of best case complexity of insertion sort?
- a) $O(n)$ b) $O(n \log n)$ c) $O(1)$ d) $O(n^2)$
- vi) How does the rear index incremented in circular queue?
- a) $\text{front} = (\text{rear} + 1) \% \text{SIZE}$ b) $\text{rear} = (\text{rear} + 1) \% \text{SIZE}$

c) $\text{rear} = \text{rear} + 1$

d) $\text{rear} = (\text{rear} - 1) \% \text{SIZE}$

vii) A variation of linked list in which none of the node contains NULL pointer is

a) Singly b) Multiple c) Circular d) Doubly viii) Which of the following data

structure is used in depth first search of graph?

a) Stack

b) Queue

c) Linked List

d) None of the above

ix) Which of the following is true for B-Tree of order M?

a) Leaf nodes should be at different level

b) All the key values within a node must be in descending order

c) Every node has at least M children

d) All non-leaf nodes with M-1 keys must have M number of children

x) Which of the following is not a hash function?

a) Division remainder

b) Folding

c) Chaining

d) Mid square



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Group B

Attempt any SIX questions.

[6 × 5 = 30]

2. What is Data Structure? Show the status of stack converting following infix expression to postfix $P + Q - (R * S / T + U) - V * W$ [1+4]
3. Write binary search. Consider a hash table of size 10; insert the keys 62, 37, 36, 44, 67, 91 and 107 using linear probing. [2+3]
4. What are deterministic and non-deterministic algorithms? Explain greedy algorithm. [3+2]
5. Draw a BST from the string DATASTRUCTURE and traverse the tree in post order and preorder. [3+2]
6. Define circular queue? How does circular queue overcome the limitation of linear queue? Explain. [2+3]
7. What is singly linked list? Write an algorithm to add a node at the beginning and end of singly linked list. [1+4]
8. Define AVL tree. Construct AVL tree from given data set: 4, 6, 12, 9, 5, 2, 13, 8, 3, 7, 11. [2+3]

Group C

Attempt any TWO questions.

[2 × 10 = 20]

9. What is stack? List the applications of stack. Write an algorithm or procedure to perform PUSH and POP operation in stack. [1+2+7]

10. What is heap? Explain quick sort algorithm with Big-oh notation in best case, average case and worst case and trace it to sort the data: 8, 10, 5, 12, 14, 5, 7, 13. [2+2+6]
11. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.



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2019

Bachelor in Computer Applications
Course Title: OOP in Java
Code No: CACS 204
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10 × 1 = 10]

1. Circle (O) the correct answer.

i) Which one of the following is not a valid java bitwise operator?

a) >> b) <<

c) >>> d) <<<

ii) Which one of the following keyword is used to declare an exception?

a) throws b) throw

c) try d) catch

iii) Which of these is an incorrect array declaration?

a) `int ary[] = new int[5];` b) `int[] ary = new int[5];`

c) `int ary = int[5] new;` d) `int ary[]; ary = new int[5];`

iv) Which one of the following access specifier is appropriate for members of superclass to access only from subclass?

a) private b) protected

c) public d) default

v) Which one of the following is not a collection class defined in java?

a) Linked List b) Hash Set

c) Tree Set

d) Graph Set

vi) Which one of the following inheritance is best implanted using interface in java?

a) single inheritance

b) multi-level inheritance

c) multiple inheritance

d) hierarchical inheritance

vii) Which one of the following method is called only once during the run time of your applet?

a) stop() b) paint()

c) init()

d) start()

viii) Which of these method of class String is used to compare two String objects for their equality?

a) equals() b) Equals()

c) is Equal()

d) Is Equal()

ix) What is the default value of priority variable MIN_PRIORITY and MAX_PRIORITY?

a) 0 & 63 b) 1 & 10

c) 0 & 1

d) 1 & 32

x) Which one of the following is not java swing container?

a) Panel b) Tabbed Pane

c) Scroll Pane

d) Scroll bar



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define OOP. Explain features of Object Oriented Programming Language. [1 + 4]
 3. Explain different types of control statements used in java. [5]
 4. Define Abstract Class. Explain different types of Access controls available in java. [1 + 4]
 5. Define method overriding? Write any program to implement concept of multiple inheritance in Java.
[1 + 4]
 6. Why it is important to handle exception in java? Write a program to illustrate the use of exception handling. [1 + 4]
 7. Define the use of **static** keyword. Write any four String methods used in java with example.
[1 + 4] 8.
- Define **super**, **final** and **this** keyword in java. Explain the concept of MVC in brief.
[1 + 1 + 1 + 2]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. a) Define multithreading. Write a java program to show the inter-thread communication.
[1 + 4]
b) Define Stream. Write a program in java to copy the content from one file to another.
[1 + 4]
10. a) Define Collection Class. Explain different Wrapper classes and associated methods in java.
[1 + 4]
b) Define AWT. Explain different types of Layout Managers in java. [1 + 4]

11. a) List and explain any five swing controls with their uses. [5]
- b) Define JDBC. Write a program to display all records from a table of database. [1 + 4]



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Bachelor in Computer Applications
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Semester: III

Full Marks: 60
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Time: 3 hours

Center:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10 × 1 = 10]

1. Circle (O) the correct answer.

i) How many types of data on the basis of sources of data collection?

a) 1 b) 2 c) 3 d) 4 ii) Which is more appropriate central tendency to

find the average of profit?

a) Arithmetic mean

b) Median

c) Mode

d) All

iii) What is the range of Correlation?

a) 0 to ∞

b) $-\infty$ to ∞

c) -1 to 1

d) 0 to 1

iv) If $r=0.2$ then coefficient of determination implies that

a) 20% of total variation in dependent variable has been explained by independent variable.

b) 40% of total variation in dependent variable has been explained by independent variable.

c) 2% of total variation in dependent variable has been explained by independent variable.

d) 4% of total variation in dependent variable has been explained by independent variable.

v) What is the minimum value of Probability?

a) 1

b) 100

c) 0

d) None of above

vi) In case of Normal distribution

a) Mean > Median

b) Mean = Median

c) Mean $\leq$ Median

d) Mean $\geq$ Median

vii) The regression line of X on Y and Y on X are intersect at the point

- a) $(\mu, 0)$ b) (a, b) c) (X, Y) d) $(\bar{X}, \bar{Y})$

viii) In case of systematic sampling

- a) sample mean is biased estimator population mean.
b) sample mean is unbiased estimator population mean.
c) sample mean can't estimate population mean.
d) sample mean may equal to population mean.

ix) Mean of Chi-Square distribution with n degrees of freedom is

- a) 1 b) 0 c) 2n d) n

x) How do you obtain degree of freedom in one-way ANOVA?

- a) $(k, n-1)$ b) $(k, n-k)$
c) $(k-1, n-1)$ d) $k-1, n-k)$



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Describe scope and limitation of Statistics.
3. Determine average wages from following data:

Wages	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

4. Calculate Karl Pearson's correlation coefficient from the following data:

Sales	43	41	36	34	50
Expenses	10	22	13	19	17

5. Estimate the marks in JAVA when the marks in Statistics is 65 by using following data:

Marks in Statistics	57	58	59	59	60	61	62	64
Marks in JAVA	77	78	75	78	82	82	79	81

6. Fit Binomial Distribution from the following data where $p = 0.5$

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

7. How do you determine sample size in sampling? Explain briefly.
8. Write short notes on simple random sampling.

Group C

Attempt any TWO questions.

[2 × 10 =

20]

9. Student's age in the regular daytime BCA program and the morning time BCA program of a campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestion, with reason, which of two groups will be easier to teach?

Regular BCA program		Morning BCA program	
Age	No. of Students	Age	No. of Students
23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	60

10. Given a normal distribution with mean 200 and s.d. 20, find the probability that

i) $P(X > 180)$

ii) $P(X < 220)$

iii) $P(160 < X < 240)$

iv) $P(X > 220)$

v) $P(X < 180 \text{ or } X > 220)$

vi) 10% of the values are less than what values of X?

11. The labor productivity indexes of Nepal are recorded as below:

Sector	Year		
	2015	2016	2017
Agriculture	100	125	138
Manufacturing	100	60	53
Community and Social service	100	89	80

Does the labor productivity index vary due to the;

- i) difference in the sector ii) difference in the time period?



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Bachelor in Computer Applications
Course Title: System Analysis & Design
Code No: CACS 203
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10 × 1 = 10]

2. Circle (O) the correct answer.

- i) Which of the following Information systems are aimed at improving the routine business activities on which all organizations depend?
 - a) Management Information systems
 - b) Decision support systems
 - c) Transaction Processing Systems
 - d) Executive Information
- ii) The project life cycle consists of
 - a) Understanding the scope of the project
 - b) Objectives of the project
 - c) Formulation and planning various activities
 - d) Word Processor
- iii) Which is the most important feature of spiral model?
 - a) Quality Management
 - b) Efficiency Management
 - c) Risk Management
 - d) Performance Management
- iv) includes the existing system, the proposed system, system flow charts, modular design of the system, print layout charts and data file designs.
 - a) Feasibility Report
 - b) Functional Specification Report
 - c) Design Specification Report
 - d) Terms of Reference

- v) For the best Software model suitable for the project, in which of the phase the developers decide a roadmap for project plan?
 - a) Software
 - b) System Analysis
 - c) Coding
 - d) Testing
- vi) Using the approach, a new system is tested in one part of the organization before being implement in others.
 - a) Direct
 - b) Parallel
 - c) phased
 - d) pilot
- vii) extends the software beyond its original functional requirements.
 - a) Adaptive maintenance
 - b) Perfective maintenance
 - c) Corrective maintenance
 - d) Preventive maintenance
- viii) Which normal form looks at removing partial dependencies?
 - a) First Normal Form
 - b) Second Normal Form
 - c) Third Normal Form
 - d) Fourth Normal Form
- ix) In constructing ER diagrams, double ovals are used to denote
 - a) Multi-value table
 - b) Multi-value entity
 - c) Multi-value attributes
 - d) Multi-value key
- x) Testing beyond normal operational capacity is
 - a) Performance testing
 - b) Stress testing
 - c) Recovery testing
 - d) None of the above



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Group B

Attempt any SIX questions.

[6 × 5 = 30]

3. When would you use agile methodologies? How is it different from waterfall approach to system development?
4. Why is project management important? Briefly explain the activities performed by the project manager during project execution.
5. List various methods of interacting with a system. Briefly explain the factors to be considered while designing a form?
6. What are the deliverables from coding and testing? Briefly explain the different approaches to installation.
7. Why is normalization required? State second normal form and explain it with a proper example.
8. Construct an E-R Diagram for football club that has a name and a ground and is made up for players. A player can play for only one club and a manager identified by his name manage a club. A footballer has a registration number, name and age. A club manager also buys players. Each club plays against other clubs in the league and matches have a date, venue and score.
9. Maintenance is an on-going process. Do you agree? Explain the process of maintaining information systems.

Group C

Attempt any TWO questions.

[2 × 10 = 20]

10. Develop a context diagram and top level logical DFD for the system made up of the following.
B & B is a mail-order company that distributes CDs, DVDs of music, games, movies, software at discount prices to club members.

- When an order processing clerk receives an order form, he or she verifies that the sender is a club member by checking the Member file.
- If the sender is not a member, the clerk returns the order along with a membership application form.
- If the customer is a member, the clerk verifies the order item data by checking the item file.
- Then the clerk enters the order data and saves it to the Daily Order file. The clerk also prints an invoice and shipping list for each order, which are forwarded to the Order Fulfillment Department.

10. With proper reasoning, explain how CASE Tools aid in information system development?
You have been hired as a system analyst in a TU tech software development company and you are asked to analyze the way the system works. What qualities do you need to have to analyze such type of systems?
11. a) Why do software projects often fail? Explain different types of software testing?
b) List of OOAD. Differentiate between structured methodologies and object-oriented methodologies.



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Bachelor of Arts in Computer Application
Course Title: Web Technology
Code No: CACS 205
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

10□1 = 10

1. Circle (O) the correct answer.

i) Which of the following provide additional information about HTML elements?

- a) Headings b) Paragraphs c) Attributes d) Styles

ii) Which of the following is NOT an inline element?

- a) `<img>` b) `<span>` c) `<input>` d) None of the above

iii) How can you make an e-mail link?

- a) `<a href="mailto:xxx@yyy">` b) `<mail href="xxx@yyy ">`
c) `<mail>xxx@yyy </mail>` d) `<a href="xxx@yyy ">`

iv) For XML document to be valid?

- a) Document need to be well formed also
b) Document need not to be well formed
c) Document need to be well formed & valid
d) Document validity has no relationship with well formedness

v) W3C is stands for

- a) World Wide Web Center
 - b) World Wide Web Company
 - c) World Wide Web Consortium
 - d) World Wide Web Certificate
- vi) XSL is,
 - a) AN XML based language to create style sheets
 - b) Extensible solution language
 - c) A language which is not used in XML
 - d) Extended stylesheet language
- vii) CSS uses dot (.) for,
 - a) Element selector
 - b) Class selector
 - c) ID selector
 - d) Attribute selector
- viii) Namespace in XML is used to,
 - a) Distinguishes one XML vocabulary from another
 - b) Provides the spaces into name
 - c) Querying language
 - d) Make duplicate of XML validation
- ix) A cookie is often used to identify a
 - a) User
 - b) Browser
 - c) Web page
 - d) Client
- x) Server side scripts is
 - a) Visible to the end user
 - b) Invisible to the end user
 - c) Defined to user
 - d) None



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Time: 3 hours

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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is CSS? Discuss types of CSS with example.
3. What is XML? Differentiate XML schema with DTD.
4. List common application of web server, Explain how session works.
5. Describe the tags and attributes for a form in HTML document?
6. What is anonymous access? Discuss about Integrated windows authentication
7. Explain tire technology with examples?
8. Write short note on following:
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 - XML DOM

Group C

Attempt any TWO questions.

[2×10= 20]

9. Differentiate between tags and attributes? Write a HTML and CSS code design your curriculum vitae (CV) using different html elements like (table, image, formatting tags, links).
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 - Address might appears multiple times
 - Address has attribute named “type” with value permanent and temporary □ Phone must be 10 digit.



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Code No: CACS 201
Semester: III

Full Marks: 60
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Name:

Symbol No:

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Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

i) What is the measurement for time complexity of an algorithm?

- | | |
|--------------------------------------|-------------------------------------|
| a) Counting microseconds | b) Counting kilobytes of algorithms |
| c) Counting number of key operations | d) Counting number of statements |

ii) Which of the following is the result of evaluation of $5\ 7\ 4 - * 8\ 4 / + ?$

- | | | | |
|------|------|-------|-------|
| a) 5 | b) 8 | c) 10 | d) 17 |
|------|------|-------|-------|

iii) What is the recursive formula for post order traversal of binary tree?

- | | |
|--------------------|--------------------|
| a) Left-Root-Right | b) Root-Left-Right |
| c) Left-Right-Root | d) Right-Left-Root |

iv) What is the number of disk movement in TOH with 4 disks?

- | | | | |
|------|-------|-------|-------|
| a) 9 | b) 14 | c) 17 | d) 15 |
|------|-------|-------|-------|

v) What is the Big-Oh of best case complexity of insertion sort?

- | | | | |
|-----------|------------------|-----------|-------------|
| a) $O(n)$ | b) $O(n \log n)$ | c) $O(1)$ | d) $O(n^2)$ |
|-----------|------------------|-----------|-------------|

vi) How does the rear index incremented in circular queue?

- | | |
|--|---|
| a) $\text{front} = (\text{rear} + 1) \% \text{SIZE}$ | b) $\text{rear} = (\text{rear} + 1) \% \text{SIZE}$ |
| c) $\text{rear} = \text{rear} + 1$ | d) $\text{rear} = (\text{rear} - 1) \% \text{SIZE}$ |

vii) A variation of linked list in which none of the node contains NULL pointer is

- | | | | |
|-----------|-------------|-------------|-----------|
| a) Singly | b) Multiple | c) Circular | d) Doubly |
|-----------|-------------|-------------|-----------|

viii) Which of the following data structure is used in depth first search of graph?

- | | | | |
|----------|----------|----------------|----------------------|
| a) Stack | b) Queue | c) Linked List | d) None of the above |
|----------|----------|----------------|----------------------|

- ix) Which of the following is true for B-Tree of order M ?
- a) Leaf nodes should be at different level
 - b) All the key values within a node must be in descending order
 - c) Every node has at least M children
 - d) All non-leaf nodes with $M-1$ keys must have M number of children
- x) Which of the following is not a hash function?
- a) Division remainder
 - b) Folding
 - c) Chaining
 - d) Mid square



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Attempt any SIX questions.

[6×5 = 30]

2. What is Data Structure? Show the status of stack converting following infix expression to post fix $P + Q - (R * S / T + U) - V * W$ [1+4]
3. Write binary search. Consider a hash table of size 10; insert the keys 62, 37, 36, 44, 67, 91 and 107 using linear probing. [2+3]
4. What are deterministic and non-deterministic algorithms? Explain greedy algorithm. [3+2]
5. Draw a BST from the string DATASTRUCTURE and traverse the tree in post order and preorder. [3+2]
6. Define circular queue? How does circular queue overcome the limitation of linear queue? Explain. [2+3]
7. What is singly linked list? Write an algorithm to add a node at the beginning and end of singly linked list. [1+4]
8. Define AVL tree. Construct AVL tree from given data set: 4, 6, 12, 9, 5, 2, 13, 8, 3, 7, 11. [2+3]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What is stack? List the applications of stack. Write an algorithm or procedure to perform PUSH and POP operation in stack. [1+2+7]
10. What is heap? Explain quick sort algorithm with Big-oh notation in best case, average case and worst case and trace it to sort the data: 8, 10, 5, 12, 14, 5, 7, 13. [2+2+6]
11. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is data structure? Explain different operations to be performed on data structure. [1 + 4]
3. Define greedy algorithm and heuristic algorithm. Briefly explain Big-Oh notation. [3+2]
4. What is circular queue? Write an algorithm to insert an item in circular queue. [2 + 3]
5. How does AVL tree differ from BST? Construct an AVL tree from following data: 35, 56, 68, 65, 44, 41, 31, 49, 20. [1+4]
6. What is B-tree? Create a B- tree of order 4 using following data: 6, 4, 22, 10, 2, 14, 3, 8, 11, 13, 5, 9. [2 + 3]
7. What is binary search? Write an algorithm to search an item using binary search. [2+3]
8. What is graph? Explain Kruskal's algorithm to construct minimum spanning tree with example. [1 + 4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Define stack. List the applications of stack. Trace the algorithm to convert infix to postfix with following infix expression $((A+B)-C*D/E)*(H-I)*F+G$ and evaluate the obtained postfix expression with following values: A=4, B=2, C=4, D=3, E=8, F=2, G=3, H=5, I=1 [1 +1+4+ 4]
10. What is double linked list? How does it differ from circular linked list? Write an algorithm or function to add a node at the beginning and end of double linked list. [1 +1+4+ 4]
11. What is heap? Differentiate between min heap and max heap. Sort the following data in ascending order by heap sort method: 2, 9, 3, 12, 15, 8, 11. [2+2+6]



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2020

Bachelor in Computer Applications
Course Title: Object Oriented Programming in Java
Code No: CACS 204
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

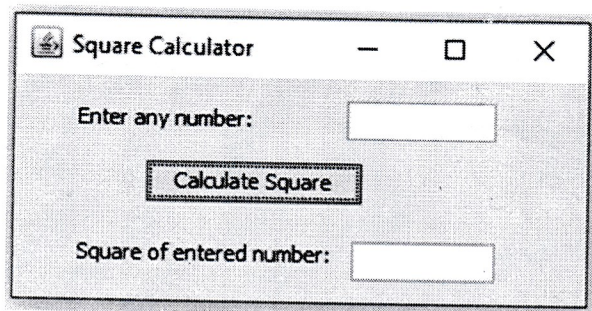
2. Define OOP. Write the characteristics of OOP Language. [1 + 4]
3. Explain the operators available in Java programming. [5]
4. Define loop. Write a java program to print first n prime numbers. [1 + 4]
5. Differentiate between abstract class and interface with suitable example. [5]
6. Define access modifier. Explain access modifiers in java with example. [1 + 4]
7. Define exception. Explain exception handling mechanism in java with example. [1+5]
8. Write short note on (Any Two): [2.5 + 2.5]
 - a) final keyword
 - b) Collection class
 - c) JDBC

Group C

Attempt any TWO questions.

[2×10 = 20]

9. a) Write a program to create and use java package. [5]
b) Define thread. Explain the life cycle of thread. [1 + 4]
10. a) Write a program to sort name of any five cities in ascending order. [5]
b) Define polymorphism. How do we achieve polymorphism in java explain with example? [1 + 5]
11. a) Differentiate between java AWT and java Swing. Explain the different types layout managers in java GUI programming. [2+3]
b) Write a java GUI program to calculate square of entered number. [5]





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Time: 3 hours

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

i) Which one of the following is not a valid java operator?

a) • (Dot)

b) >>>

c) instanceof

d) system

ii) Which one of the following keyword is not used in exception handling?

a) throws

b) throw

c) try

d) tries

iii) Which one of the following is not a valid jumping statement in Java?

a) continue

b) break

c) goto

d) return

iv) Which one of the following access specifier is appropriate for members of superclass to access only from subclass?

a) private

b) protected

c) public

d) default

v) Which one of the following is not a collection class defined in java?

a) Linked List

b) Hash Set

c) Tree Set

d) Graph Set

vi) Which one of the following inheritance uses interface?

a) single inheritance

b) multi-level inheritance

c) multiple inheritance

d) hierarchical inheritance

vii) Which one of the following method is called only once during the run time of your applet?

a) stop()

b) paint()

c) init()

d) start()

viii) Which of these method of class String is used to compare two String objects for their equality?

a) equals()

b) Equals()

c) is Equal()

d) Is Equal()

ix) What is the default value of priority variable MIN_PRIORITY and MAX_PRIORITY?

a) 0 & 63

b) 1 & 10

c) 0 & 1

d) 1 & 32

x) Which one of the following is not java swing menu control?

a) Menu Bar

b) Menu

c) Menu Item

d) Menu Scroll



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define OOP. Explain features of Object Oriented Programming Language. [1 + 4]
3. Explain different types of looping statements used in java. [5]
4. Define Abstract Class. Explain different types of Access controls available in java. [1 + 4]
5. How polymorphism is achieved in Java? Explain with example. [1 + 4]
6. Define Exception Handling. Write a program to illustrate the use of exception handling. [1 + 4]
7. Define the use of **static** keyword. Write any four String methods used in java with example. [1 + 4]
8. Define **super**, **final** and **this** keyword in java. Explain the concept of MVC in brief. [1+1+1+2]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. a) Define multithreading. Write a java program to show the inter-thread communication. [1+4]
b) Define Stream. Write a program in java to copy the content from one file to another. [1+4]
10. a) Define Collection Class. Explain different Wrapper classes in java.
[1 + 4]
b) Define AWT. Explain different types of Layout Managers in java. [1 + 4]
11. a) List and explain any five swing controls with their uses. [5]
b) Define JDBC. Write a program to display all records from a table of database. [1 + 4]



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Bachelor in Computer Applications
Course Title: Probability and Statistics
Code No: CAST 202
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Name:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

i) How many types of data on the basis of sources of data collection?

- a) 1 b) 2 c) 3 d) 4

ii) Which is more appropriate central tendency to find the average of profit?

- a) Arithmetic mean b) Median
c) Mode d) All

iii) What is the range of Correlation?

- a) 0 to ∞ b) $-\infty$ to ∞ c) -1 to 1 d) 0 to 1

iv) If $r=0.2$ then coefficient of determination implies that

- a) 20% of total variation in dependent variable has been explained by independent variable.
b) 40% of total variation in dependent variable has been explained by independent variable.
c) 2% of total variation in dependent variable has been explained by independent variable.
d) 4% of total variation in dependent variable has been explained by independent variable.

v) What is the minimum value of Probability?

- a) 1 b) 100 c) 0 d) None of above

vi) In case of Normal distribution

- a) Mean > Median b) Mean = Median c) Mean $\leq$ Median d) Mean $\geq$ Median

vii) The regression line of X on Y and Y on X are intersect at the point

- a) $(\mu, 0)$ b) (a, b) c) (X, Y) d) $(\bar{X}, \bar{Y})$

viii) In case of systematic sampling

- a) sample mean is biased estimator population mean.
b) sample mean is unbiased estimator population mean.
c) sample mean can't estimate population mean.
d) sample mean may equal to population mean.

ix) Mean of Chi-Square distribution with n degrees of freedom is

- a) 1 b) 0 c) 2n d) n

x) How do you obtain degree of freedom in one-way ANOVA?

- a) $(k, n-1)$ b) $(k, n-k)$
c) $(k-1, n-1)$ d) $k-1, n-k)$



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Describe scope and limitation of Statistics.
3. Determine average wages from following data:

Wages	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

4. Calculate Karl Pearson's correlation coefficient from the following data:

Sales	43	41	36	34	50
Expenses	10	22	13	19	17

5. Estimate the marks in JAVA when the marks in Statistics is 65 by using following data:

Marks in Statistics	57	58	59	59	60	61	62	64
Marks in JAVA	77	78	75	78	82	82	79	81

6. Fit Binomial Distribution from the following data where $p = 0.5$

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

7. How do you determine sample size in sampling? Explain briefly.
8. Write short notes on simple random sampling.

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Student's age in the regular daytime BCA program and the morning time BCA program of a campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestion, with reason, which of two groups will be easier to teach?

Regular BCA program		Morning BCA program	
Age	No. of Students	Age	No. of Students
23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	60

10. Given a normal distribution with mean 200 and s.d. 20, find the probability that
- i) $P(X > 180)$
 - ii) $P(X < 220)$
 - iii) $P(160 < X < 240)$
 - iv) $P(X > 220)$
 - v) $P(X < 180 \text{ or } X > 220)$
 - vi) 10% of the values are less than what values of X?
11. The labor productivity indexes of Nepal are recorded as below:

Sector	Year		
	2015	2016	2017
Agriculture	100	125	138
Manufacturing	100	60	53
Community and Social service	100	89	80

Does the labor productivity index vary due to the;

- i) difference in the sector
- ii) difference in the time period?



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. Write down the process of collecting primary data.
3. Determine First Quartile (Q_1) 7th Decile (D_7) and 80th Percentile (P_{80}) from the following data:

Age in year	10	12	14	16	18	20	22	24	26
No. of people	7	11	24	35	27	17	11	8	5

4. Calculate correlation coefficient between income and expenditure in foods of certain families of Kathmandu Metropolitan from the following information:

Income (000Rs)	10	11	12	13	14
Expenditure in foods (000Rs)	9	8	9	12	11

5. A box contains 50 items of which 20 are defectives. If one item is selected randomly from the box, what is the probability that it is a non-defective item?
6. What is sampling? The standard deviation of marks in an entrance exam of BCA students is 0.5. How large a sample must be taken in order to be 95% confidence that the error of his/her estimate will not exceed 0.01.
7. Calculate the median and mode from following distribution:

Expenditure(000Rs)	10-20	20-30	30-40	40-50	50-60	60-70
Number of families	7	12	15	13	8	5

8. A test was given to three candidates taken at random from three provinces of Nepal. The scores of candidates are given below:

Gandaki	9	7	6
Lumbini	7	4	5
Bagmati	6	5	6

Carry out one-way ANOVA.

Group C

Attempt any TWO questions.

[2×10 = 20]

9. From the following data, determine average marks of student, standard deviation and coefficient variation.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of people	54	90	86	58	62	82	78	66	70

10. In a normal distribution with mean = 200 and standard deviation = 20, find the probability that
a) $P(X > 180)$ b) $P(X < 220)$ c) $P(160 < X < 240)$ d) $P(X > 220)$
e) 10% values are less than what values of X?
11. Describe simple random sampling with a suitable example.



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Bachelor in Computer Applications
Course Title: System Analysis & Design
Code No: CACS 203
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Name:	Symbol No:
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Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

- i) is an important factor of Management Information System.
 - a) System
 - b) Data
 - c) Process
 - d) All
- ii) In a DFD external entities are represented by a
 - a) rectangle
 - b) ellipse
 - c) diamond
 - d) circle
- iii) After the design phase the document prepared is known as
 - a) system specification
 - b) performance
 - c) design specification
 - d) none of the above
- iv) can be defined as most recent and perhaps most comprehensive technique for solving computer problems.
 - a) System Analysis
 - b) System Data
 - c) System Procedure
 - d) System record

- v) SDLC stands for
- a) System Development Life Cycle
 - b) System Design Life Cycle
 - c) Structure Design Life Cycle
 - d) Structure Development Life Cycle
- vi) is a tabular method for describing the logic of the decision as to be taken
- a) Decision tables
 - b) Decision tree
 - c) Decision method
 - d) Decision Data
- vii) Documentation is prepared
- a) at every stage
 - b) at system design
 - c) at system analysis
 - d) at system development
- viii) Which of the following is not a characteristic of good test data?
- a) Users do not participate at this preliminary stage.
 - b) Should be comprehensive.
 - c) Every statement should be executed.
 - d) All of the above.
- ix) System approval criteria are specified
- a) When the first specification is drawn up.
 - b) During feasibility study
 - c) During the requirement specification state
 - d) During system study state
- x) JAD stands for
- a) Just-in Application Design
 - b) Joint Application Design
 - c) Joint Application Development
 - d) Joint Applied Design



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Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Elaborate on the steps in feasibility analysis.
3. Explain what is system and what are its characteristics of a system in details?
4. Why is it difficult to determine user requirements? Illustrate the key strategies for eliciting information about the user's requirements.
5. Distinguish between system and user documentation and determine which types of documentation are necessary for a given Information system.
6. Explain and contrast four types of maintenance.
7. Differentiate between Questionnaire and interview survey.
8. What is difference between a 2 NF & 3 NF relation?

Group C

Attempt any TWO questions.

[2×10 = 20]

9. List and describe the skills and activities of a project manager during project initiation, project execution and project close down.
10. Draw context diagram, Data flow Diagram up to level-2 of food ordering system of Bakery Café Jawala Khel- Lalitpur.
11. How can you transform E-R diagram into relations? Explain with suitable examples. A Bank has the following policy on deposits of RS. 50,000/- and above for five years or above, the interest rate is 15%. On the same deposit for period users than 5 years it is 12%. On the deposit below RS. 50,000/-, the interest rate is 10% regardless of period and deposit. Write the above process using

a) Decision Table

b) Decision Tree



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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is system? What are the phases of SDLC? Explain briefly. [1+4]
3. What do you mean by planning? Write the process of planning for Information System Development Project. [1+4]
4. Define process modeling. Explain DFD with example. [1+4]
5. List and explain the skills and responsibilities of project manager. [2.5+2.5]
6. Explain the guidelines to design in an interface and dialogue box for e-commerce system. [5]
7. Differentiate between system and user documentation with their applications. [5]
8. Define software testing. Explain software quality assurance activities. [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What are the major differences between Agile methodologies and waterfall model? Why should you use agile methodologies? Explain. [5+5]
10. How can you transform ER Diagram into relation? Explain with your own suitable example. [10]
11. Why is the project management important? Describe the concept of integrated CASE tools with its applications. [3+7]



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Bachelor in Computer Applications
Course Title: Web Technology
Code No: CACS 205
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Name:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt **all** the questions.

[10 x 1 = 10]

1. Circle (O) the correct answer.

i) Why we used XSLT language?

- a) Use to transport XML documents
- b) use to perform transaction b/w XML documents
- c) Use to transform XML documents
- d) use to format XML documents

ii) Which is the correct CSS syntax?

- a) body {color: black}
- b) {body;color:black}
- c) {body:color=black(body)}
- d) body:color=black

iii) SQL Query to delete all rows in a table without deleting the table structure.

- a) DELETE FROM table_name;
- b) DELETE TABLE table_name;
- c) DROP TABLE table_name;
- d) NONE

iv) Web design that makes your web page look good on all devices is called as?

- a) Good web design
- b) Interactive web design
- c) Responsive web design
- d) Both A and B

v) What is the constraint on the data per cookie?

- a) 2 KB
- b) 1 KB
- c) 4 KB
- d) 3 KB

vi) Which of following is not an inline element?

- a)
- b) <a>
- c)
- d) <div>

vii) Which is wrong in XQuery?

- a) Used for transforming XML data into XHTML
- b) Used for search web documents
- c) Used to generate tables for XSLT
- d) None of these

viii) Which method has much larger limit on amount of data that can be passed to web server?

- a) Get method b) Post method c) Put method d) Set method

ix) In HTML form <input type="text"> is used for

- a) One-line text b) Block of text c) One paragraph d) None

x) Which inline function embeds an independent HTML document into current document?

- a) <div> b) c) <iframe> d) <form>



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Group B

Attempt any **SIX** questions.

[6 x 5 = 30]

2. What is HTML Link? Explain different link used in HTML.

[1+4]

3. Write HTML tag to generate the following table.

[5]

Routine			
Sunday	Monday	Break	Tuesday
WT	DSA		SAD
Java	Statistics		Math

4. Write HTML tag to generate the following output.

[5]

○ FOHSS

▪ BCA

1. IoT

▪ BSW

○ FOM

1. BIM

2. BBA

5. “It is discouraged to use frame in web development” Explain the reasons.

[5]

6. What is the strength of CSS? What are the various approaches to include CSS document in HTML?

[1+4]

7. What is cookie? Explain how you set and remove values stored in cookie with suitable example?

[1+4]

8. Write HTML tag to generate the following output.

[5]

Student Information

First Name

Last Name

Gender

☒ Male ☐ Female ☐ Other

Subject

☐ C ☒ JAVA ☐ PHP

Comment

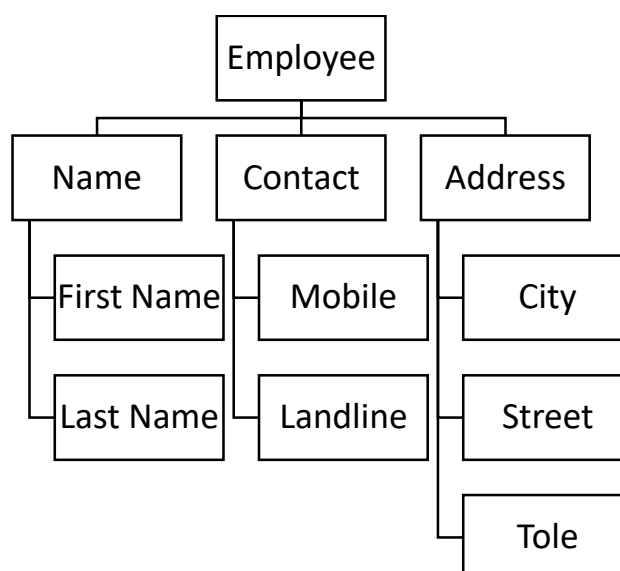
Submit

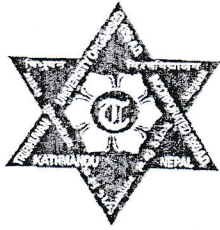
Group C

Attempt any **TWO** questions

[2 x 10 = 20]

9. Is XML replacement for HTML? Explain different types of tier technology with suitable diagram. [2+8]
10. Write a server-side script to display all the records store in student table which resides under the database called TU. The structure of student table as (id, name, address). [10]
11. Write well-formed XML document and validate with the use of XML schema. [5+5]





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Group B

Attempt any SIX questions.

[6×5 = 30]

2. What is HTML5? Explain characteristics of HTML5. [5]

3. Write HTML tag to generate the following table. [5]

	Average		Red
	Height	Weight	Eyes
Males	1.5	0.004	45%
Females	1.4	0.006	47%

4. Design the following website's Menu with use of HTML list and appropriate CSS. [5]

Home | News | Contact | About

5. Compare and contrast between a block-level element and an inline element in HTML? [5]

6. What is CSS Selector? Explain different types of selector with suitable example. [1+4]

7. Why session is required in web development? Explain how you set and remove values stored in Session with suitable example? [1+4]

8. Critically analysis the pitfall of the 3-tier technology in comparison with n-tier technology.

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Design the following layout of webpage with use of div and appropriate CSS. [10]

Logo	
Navigation	
Header Banner	
Side Bar	Body Area
Footer	

10. Write a server-side script for login process assume that user name and password have already exist on database TU under the **user** table (id, **user**, **password**). [10]

Login Form

User Name

Password

[Lost your password?](#)

Don't have an account? [Signup here!](#)

11. What are the rules for Well-formed xml document? Create a sample well-formed xml and validate it with DTD Schema. [5+5]